Homework3

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Question1: Solution:
Shape p_i depends only on the class
Rate λ_j only on the feature.

a) Maximize $g_i(\mathbf{x}) = \ln p(\mathbf{x} \mid \omega_i) + \ln P(\omega_i)$.
We get:

$$g_i(\mathbf{x}) = p_i \sum_{j=1}^{k} \ln(\lambda_j x_j) - k \ln \Gamma(p_i) + \ln P(\omega_i)$$

b) $g_i - g_l = (p_i - p_l) \sum_j \ln(\lambda_j x_j) + consistence.$
x-dependence is $(p_i - p_l) \sum_j \ln x_j$ which linear in $\ln x_j$, don't linear in x_j.
It vanishes only when $p_i = p_l$, butthe densities are identical
No boundary. So the boundary is never linear in the original x.

c) p_1=4, p_2=2 ($\lambda = (1, 2, 1, 2)$)
Then,
$T = \sum_j \ln(\lambda_j x_j)$
$= \ln 0.1 + \ln 0.4 + \ln 0.3 + \ln 8 = -2.343$

$$g_1 = 4T - 4 \ln 6 = -16.54, \quad g_2 = 2T - 4 \ln 1 = -4.69$$

Then we get ω_2.

d) k=1, λ=1, p_1=3.2, p_2=8
With equal densities:

$$x^{p_2 - p_1} = \frac{\Gamma(8)}{\Gamma(3.2)} = \frac{5040}{2.424} = 2079 x^* = 2079^{1/4.8} \approx 4.91.$$

Decide ω_1 if $x < x^*$, else ω_2.
With ω_1 as null:

$$Type - 1 = Q(3.2, x^*) \approx 0.157 \qquad Type - 2 = P(8, x^*) \approx 0.124$$

(P, Q are the regularized lower/upper incomplete gamma)
total error ≈ 0.141.

e) p_1=p_2=4 and shared rates
We get: p(x| ω_1) $\equiv p(\mathbf{x} \mid \omega_2)$.
Then

$$f(x_1, x_2) = g_1 - g_2 = \ln 1/43/4 = -\ln 3 \approx -1.099 \neq 0.$$

No boundary; assign all of space to ω_2

Question2: Solution:
g$_j(\mathbf{x}) = \ln\left(\prod_{i=1}^{k} \frac{\lambda_i}{2} e^{-\lambda_i |x_i - m_{ij}|}\right) = \sum_{i=1}^{k} \ln\left(\frac{\lambda_i}{2}\right) - \sum_{i=1}^{k} \lambda_i |x_i - m_{ij}|$
Drop class-independent term $\sum \ln(\lambda_i/2)$:
$\arg\max_j g_j(\mathbf{x}) = \arg\min_j \sum_{i=1}^{k} \lambda_i |x_i - m_{ij}|$
$(Weighted Manhattan)$
for standard Manhattan distance: $\lambda_1 = \lambda_2 = \ldots = \lambda_k = \lambda$

Question3: Solution:
Joint and Marginal Probabilities
$P(x, \omega_j) = p(x|\omega_j)P(\omega_j)$ for each x and ω_j.
For $x = 1$:
$P(1, \omega) = [\frac{10}{300}, \frac{30}{300}, \frac{25}{300}, \frac{24}{300}]$.
Marginal $p(1) = \sum_j P(1, \omega_j) = \frac{89}{300}$.
For $x = 2$:
$P(2, \omega) = [\frac{10}{300}, \frac{15}{300}, \frac{50}{300}, \frac{24}{300}]$.
Marginal $p(2) = \sum_j P(2, \omega_j) = \frac{99}{300}$.
For $x = 3$:
$P(3, \omega) = [\frac{10}{300}, \frac{15}{300}, \frac{75}{300}, \frac{12}{300}]$.
Marginal $p(3) = \sum_j P(3, \omega_j) = \frac{112}{300}$.
a) Conditional Risk $R(\alpha_i|x)$ $R(\alpha_i|x) = \sum_{j=1}^{4} \lambda(\alpha_i|\omega_j)P(\omega_j|x)$.
Using $P(\omega_j|x) = \frac{P(x,\omega_j)}{p(x)}$;
$x = 1$: R$(\alpha_1|1) = [0(10) + 2(30) + 3(25) + 4(24)]/89 = 231/89 \approx 2.596$
$R(\alpha_2|1) = [1(10) + 0(30) + 1(25) + 8(24)]/89 = 227/89 \approx 2.551$
$R(\alpha_3|1) = [3(10) + 2(30) + 0(25) + 2(24)]/89 = 138/89 \approx 1.551 \quad (Min)$
$R(\alpha_4|1) = [5(10) + 3(30) + 1(25) + 0(24)]/89 = 165/89 \approx 1.854$
$x = 2$: R$(\alpha_1|2) = [0(10) + 2(15) + 3(50) + 4(24)]/99 = 276/99 \approx 2.788$
$R(\alpha_2|2) = [1(10) + 0(15) + 1(50) + 8(24)]/99 = 252/99 \approx 2.545$
$R(\alpha_3|2) = [3(10) + 2(15) + 0(50) + 2(24)]/99 = 108/99 \approx 1.091 \quad (Min)$
$R(\alpha_4|2) = [5(10) + 3(15) + 1(50) + 0(24)]/99 = 145/99 \approx 1.465$
$x = 3$: R$(\alpha_1|3) = [0(10) + 2(15) + 3(75) + 4(12)]/112 = 303/112 \approx 2.705$
$R(\alpha_2|3) = [1(10) + 0(15) + 1(75) + 8(12)]/112 = 181/112 \approx 1.616$
$R(\alpha_3|3) = [3(10) + 2(15) + 0(75) + 2(12)]/112 = 84/112 = 0.750 \quad (Min)$
$R(\alpha_4|3) = [5(10) + 3(15) + 1(75) + 0(12)]/112 = 170/112 \approx 1.518$
$\alpha(x_i)$**:** For all observed values of x, the optimal decision that minimizes the conditional risk is α_3.
b) Overall Risk R
R is the expected value of the minimum conditional risks;
R $= \sum_{i=1}^{3} R(\alpha(x_i)|x_i)p(x_i)$
$= R(\alpha_3|1)p(1) + R(\alpha_3|2)p(2) + R(\alpha_3|3)p(3)$

$= \left(\frac{138}{89} \times \frac{89}{300}\right) + \left(\frac{108}{99} \times \frac{99}{300}\right) + \left(\frac{84}{112} \times \frac{112}{300}\right)$
$= \frac{138+108+84}{300}$
$= \frac{330}{300}$
$= 1.1$
The overall risk R is 1.1.

Question4: Solution:

Supposing ω_1 represent the class "Evade = No"
ω_2 represent the class "Evade = Yes".
The irrelevant feature is "Tax ID". The total number of instances is $N = 10$.
The class counts are $N_1 = 7$ and $N_2 = 3$.

a) Estimate prior class probabilities
$\mathrm{P}(\omega_1) = \frac{N_1}{N} = \frac{7}{10} = 0.7$
$P(\omega_2) = \frac{N_2}{N} = \frac{3}{10} = 0.3$

b) Continuous feature conditional pdfs
Using Maximum Likelihood Estimates (MLE): $\hat{\mu}_i = \frac{1}{N_i}\sum_{k=1}^{N_i} x_k, \quad \hat{\sigma}_i^2 = \frac{1}{N_i}\sum_{k=1}^{N_i}(x_k - \hat{\mu}_i)^2$

For ω_1:
$X_{inc} = \{122, 77, 106, 210, 72, 117, 60\}$ $\hat{\mu}_1 = \frac{764}{7} \approx 109.14K$
$\hat{\sigma}_1^2 = \frac{1}{7}\left[(122 - 109.14)^2 + \ldots + (60 - 109.14)^2\right] \approx 2176.69$
$p(x_{inc}|\omega_1) = \frac{1}{\sqrt{2\pi(2176.69)}} \exp\left(-\frac{(x_{inc} - 109.14)^2}{2(2176.69)}\right)$

For ω_2:
$X_{inc} = \{88, 90, 85\}$ $\hat{\mu}_2 = \frac{263}{3} \approx 87.67K$
$\hat{\sigma}_2^2 = \frac{1}{3}\left[(88 - 87.67)^2 + (90 - 87.67)^2 + (85 - 87.67)^2\right] \approx 4.22$
$p(x_{inc}|\omega_2) = \frac{1}{\sqrt{2\pi(4.22)}} \exp\left(-\frac{(x_{inc} - 87.67)^2}{2(4.22)}\right)$

c) Discrete PMF estimation
(X_{ref}): $\mathrm{P}(\mathrm{X}_{ref} = Yes|\omega_1) = 3/7, \quad P(X_{ref} = No|\omega_1) = 4/7$
$P(X_{ref} = Yes|\omega_2) = 0/3 = 0, \quad P(X_{ref} = No|\omega_2) = 3/3 = 1$
Marital Status (X_{mar}):
$\mathrm{P}(\mathrm{X}_{mar} = Single|\omega_1) = 2/7, \quad P(X_{mar} = Married|\omega_1) = 4/7, \quad P(X_{mar} = Divorced|\omega_1) = 1/7$
$P(X_{mar} = Single|\omega_2) = 2/3, \quad P(X_{mar} = Married|\omega_2) = 0/3 = 0, \quad P(X_{mar} = Divorced|\omega_2) = 1/3$
Question: A valid estimate of the PMF?
Yes, because for each feature and given class, the probabilities are non-negative and sum to 1.

d) Issue & Laplace correction
Issue: In Naive Bayes, probabilities are calculated via multiplication.
If a feature category never appears in the training data, its estimated probability is 0. Consequently, the entire joint probability becomes 0.

Laplace Correction: To prevent this issue, we add a pseudo-count of 1 to the numerator and the number of feature levels to the denominator. By assigning a small and non-zero baseline probability to unseen features could ensure that no single feature can eliminate a class.

Question:A valid estimate?
Yes, the probabilities over all possible levels still sum to 1:
$\sum_{j=1}^{l} \frac{n_{ji}+1}{n_i+l} = \frac{\sum_{j=1}^{l} n_{ji}+l}{n_i+l} = \frac{n_i+l}{n_i+l} = 1$

e) Compute the Laplace-corrected probabilities:
$l = 2$:
$P_{Lap}(Yes|\omega_1) = \frac{3+1}{7+2} = \frac{4}{9}, \quad P_{Lap}(No|\omega_1) = \frac{4+1}{7+2} = \frac{5}{9}$
$P_{Lap}(Yes|\omega_2) = \frac{0+1}{3+2} = \frac{1}{5}, \quad P_{Lap}(No|\omega_2) = \frac{3+1}{3+2} = \frac{4}{5}$
Marital Status ($l = 3$):
$P_{Lap}(Single|\omega_1) = \frac{2+1}{7+3} = \frac{3}{10}, \quad P_{Lap}(Married|\omega_1) = \frac{4+1}{7+3} = \frac{5}{10}, \quad P_{Lap}(Divorced|\omega_1) = \frac{1+1}{7+3} = \frac{2}{10}$
$P_{Lap}(Single|\omega_2) = \frac{2+1}{3+3} = \frac{3}{6}, \quad P_{Lap}(Married|\omega_2) = \frac{0+1}{3+3} = \frac{1}{6}, \quad P_{Lap}(Divorced|\omega_2) = \frac{1+1}{3+3} = \frac{2}{6}$

Calculate:
$g_1(\mathbf{x}) = 0.7 \cdot P_{Lap}(x_{ref}|\omega_1) \cdot P_{Lap}(x_{mar}|\omega_1) \cdot \frac{1}{\sqrt{2\pi(2176.69)}} \exp\left(-\frac{(x_{inc}-109.14)^2}{2(2176.69)}\right)$
$g_2(\mathbf{x}) = 0.3 \cdot P_{Lap}(x_{ref}|\omega_2) \cdot P_{Lap}(x_{mar}|\omega_2) \cdot \frac{1}{\sqrt{2\pi(4.22)}} \exp\left(-\frac{(x_{inc}-87.67)^2}{2(4.22)}\right)$
Classify as Evade = Yes (ω_2) if $g_2(\mathbf{x}) > g_1(\mathbf{x})$,
otherwise **classify as Evade = No (ω_1).**

Question5: Solution:
a) Download the WPBC data
Note: the dataset is uploaded to Google drive and processed in Colab.
Using GaussianNB from sklearn.
b) As the Question b) said :Select the first 130 non-recurrent cases and the first 37 recurrent cases as your training set. Add record 197 in the data set to your training set as well.
Results shown below:
train:
Outcome
0 130
1 37
Name: count, dtype: int64
test:
Outcome
0 21
1 10
Name: count, dtype: int64
c)Replacing the missing features.
Results are shown in the Table 1.
Calculated Training Set Median: 1.0

d)i Base Naïve Bayes
Results are shown in the Table 2&3.

d)ii Base Naïve Bayes (SMOTE + Downsampling)
Results are shown in the Table 4&5.

Table 1: Training Set Instances with Missing *Lymph_Node_Status*

Python Index	**Patient ID**	**Outcome**	**Tumor_Size**	**Lymph_Node_Status**
6	844359	0	1.5	?
28	854253	0	1.5	?
85	877500	0	1.5	?
196	947204	1	3.0	?

Table 2: Classification metrics: train vs. test

Set	Precision	Recall	F1-Score	AUC
Train	0.3542	0.4595	0.4000	0.6736
Test	0.3000	0.3000	0.3000	0.5429

Table 3: Confusion matrices (rows = actual, columns = predicted)

Train

		Predicted	
		0	1
2*Actual	0	99	31
	1	20	17

Test

		Predicted	
		0	1
2*Actual	0	14	7
	1	7	3

Table 4: Classification metrics: train vs. test

Set	Precision	Recall	F1-Score	AUC
Train	0.6628	0.6333	0.6477	0.7440
Test	0.3333	0.4000	0.3636	0.6476

Table 5: Confusion matrices (rows = actual, columns = predicted)

Train

		Predicted	
		0	1
2*Actual	0	61	29
	1	33	57

Test

		Predicted	
		0	1
2*Actual	0	13	8
	1	6	4